

Orientation error by adapter, averaged over every encoder and all three training approaches

| adapter                 | median yaw | mean yaw | best yaw | success | $\Delta$ vs spatial softmax | encoders it is registered on                    |
|-------------------------|------------|----------|----------|---------|-----------------------------|-------------------------------------------------|
| Attention pooling (AFA) | 20°        | 35°      | 6°       | 0.38    | -23°                        | DINOv2, R3M L3, R3M L4                          |
| Spatial softmax         | 72°        | 53°      | 8°       | 0.21    | reference                   | Nature CNN, Compact ViT, DINOv2, R3M L3, R3M L4 |
| Local grid              | 86°        | 63°      | 22°      | 0.11    | +2°                         | DINOv2, R3M L3, R3M L4                          |
| Flatten                 | 89°        | 74°      | 6°       | 0.14    | +32°                        | Nature CNN, Compact ViT                         |
| Linear (global pool)    | 87°        | 84°      | 47°      | 0.01    | +23°                        | DINOv2, R3M L4                                  |